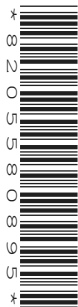


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0680/22

May/June 2018

1 hour 45 minutes

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

DO **NOT** WRITE IN ANY BARCODES.

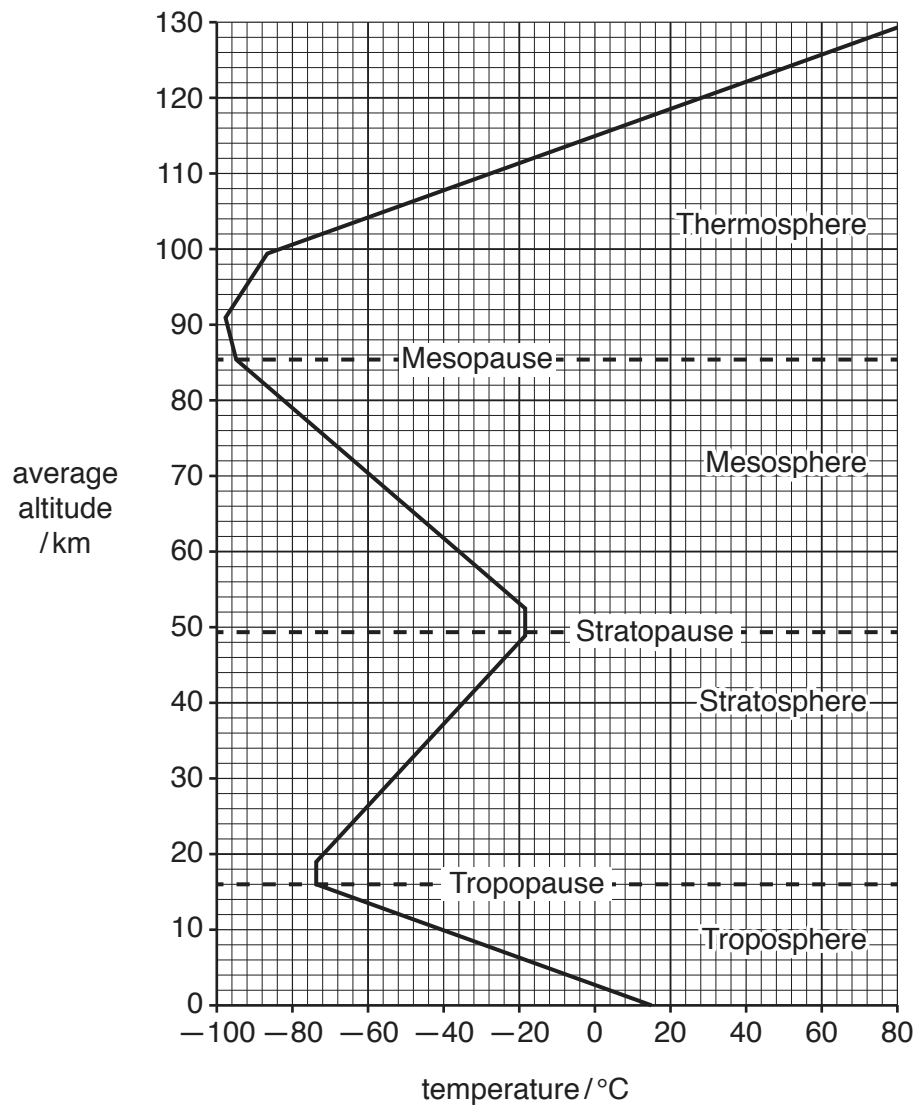
Answer **both** questions.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **14** printed pages and **2** blank pages.

- 1 (a) The diagram shows the structure of the atmosphere.



- (i) State the average altitude and air temperature at the tropopause.

altitude km temperature °C [2]

- (ii) Describe the changes in temperature above the troposphere.

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..... [3]

(iii) The ozone layer occurs at an altitude of approximately 20 to 30 km. Draw the ozone layer on the diagram of the structure of the atmosphere. [1]

(iv) Explain why the ozone layer is vital to life on Earth.

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..... [2]

(v) Above Antarctica there is an area of reduced ozone concentration, commonly known as the ozone hole.

Explain how the damage to the ozone layer over Antarctica has occurred.

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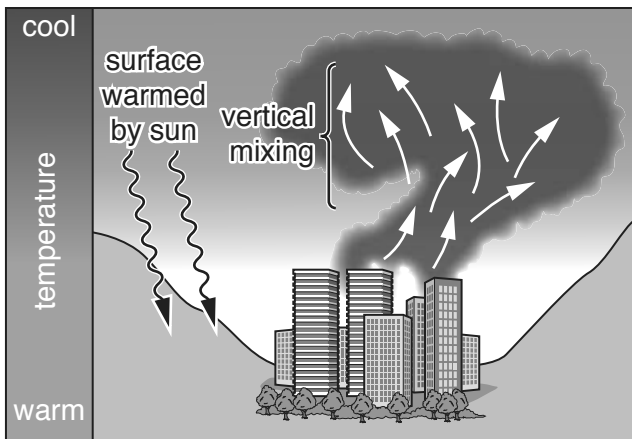
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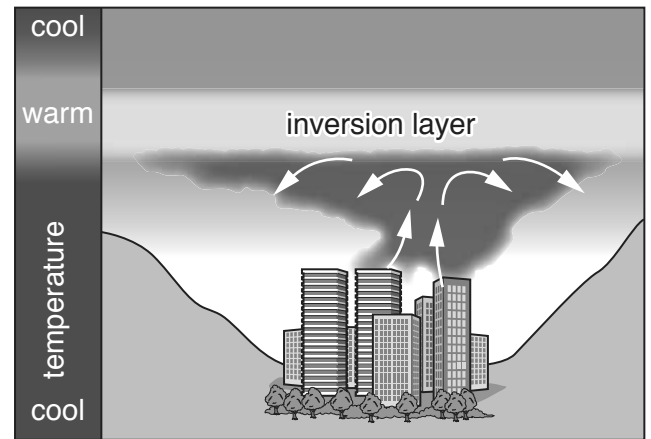
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..... [4]

(b) The diagrams show normal atmospheric conditions and a temperature inversion.



normal atmospheric conditions



temperature inversion

- (i) Using the diagrams, describe why atmospheric temperatures are usually warmest near the ground surface.

.....
 [1]

- (ii) Using the diagrams, describe what is meant by a *temperature inversion*.

.....

 [2]

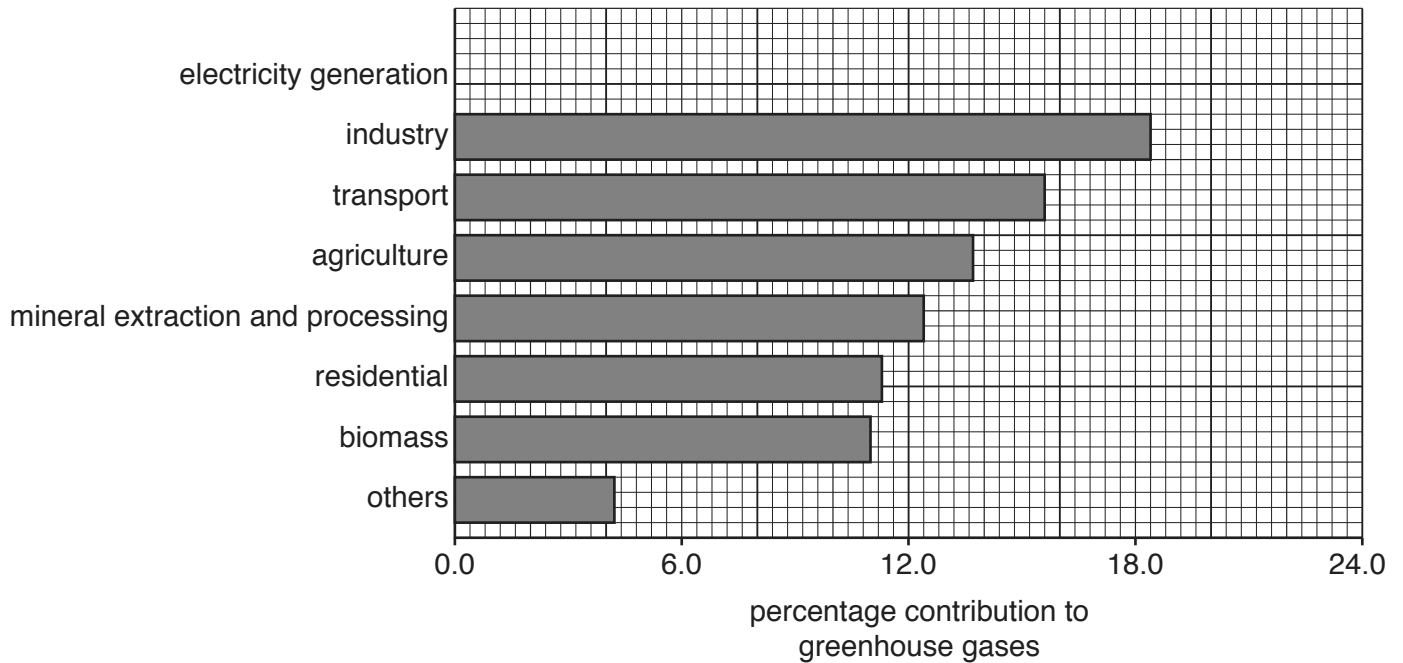
- (iii) Explain how a temperature inversion increases atmospheric pollution in a city.

.....

 [3]

(c) The graph shows sources of greenhouse gases in the atmosphere.

sources of greenhouse gas



(i) Complete the graph by showing an electricity generation contribution to greenhouse gases of 21.0%. [1]

(ii) State the percentage contribution to greenhouse gases from transport.

.....% [1]

(iii) Explain how electricity generation and agriculture produce greenhouse gases.

electricity generation

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.....

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agriculture

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[5]

(iv) Suggest ways in which greenhouse gas emissions can be reduced.

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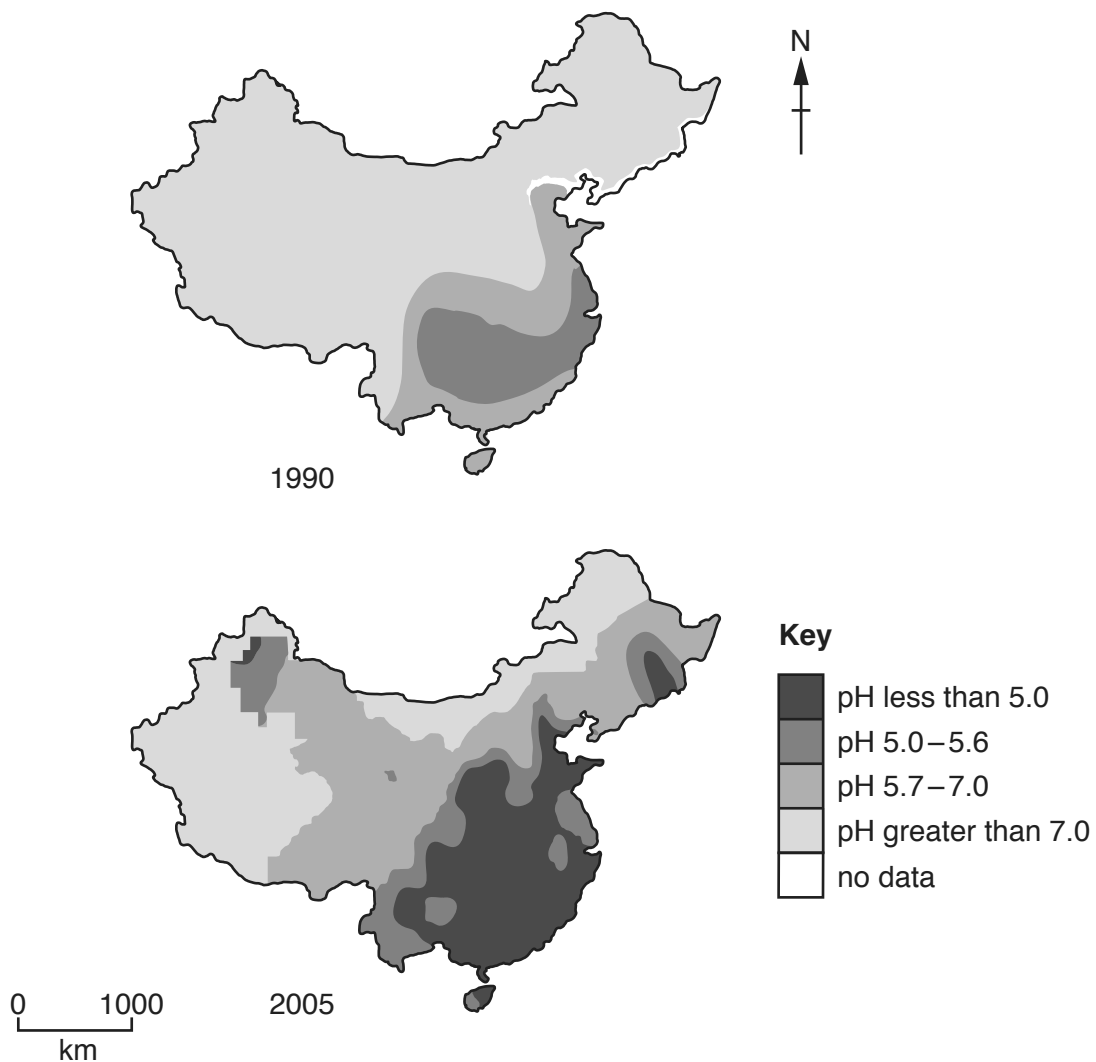
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..... [3]

(d) The maps show the pH of rainfall in China in 1990 and 2005. The lower the pH, the more acidic the rainfall.



(i) State the lowest pH of rainfall in China in 1990.

..... [1]

- (ii) Describe the changes in the pH of rainfall in China from 1990 to 2005.

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..... [3]

- (iii) Suggest **two** reasons for the changes in the pH of rainfall in China from 1990 to 2005.

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..... [2]

- (e) Is international cooperation necessary to overcome the problems of atmospheric pollution?
Give reasons for your answer.

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..... [6]

- 2 (a) The newspaper article is about the link between toxins (pollutants) in the oceans and the death of whales.

Increase in whale deaths around the UK

Scientists think that flame retardants and polychlorinated biphenyls (PCBs) released in the oceans damage the immune system of whales. Whales then become infected with diseases and parasites which cause them to become confused and swim onto beaches where they die. Whales accumulate toxins from the food they eat. Almost 600 whales died on UK beaches in 2014. It has been found that these whales had the highest concentration of PCBs in their bodies of whales anywhere in the world.

- (i) State the **two** toxins from the article thought to cause the death of whales.

..... and [1]

- (ii) Explain why toxin levels are higher in top predators than in animals lower in the food chain.

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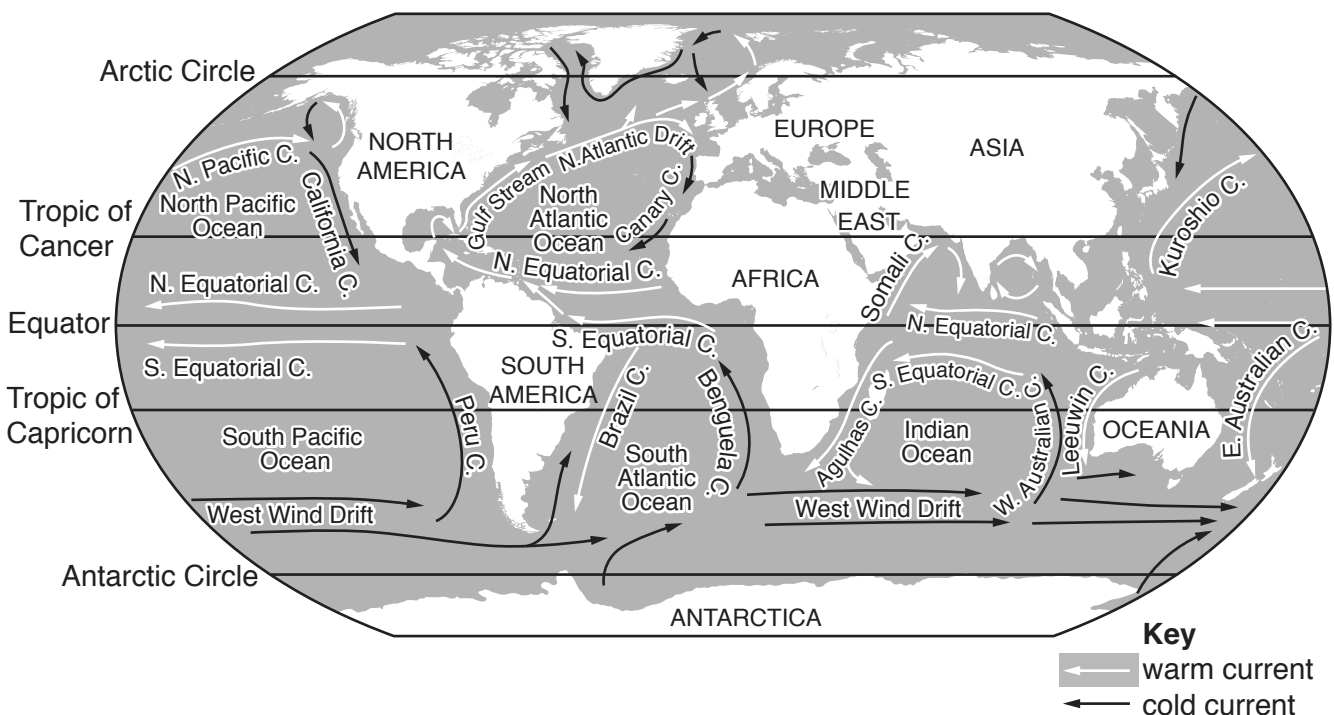
 [2]

- (iii) Suggest how toxins enter the oceans.

.....

 [2]

- (b) The map shows warm and cold ocean currents.



- (i) Name the cold current along the west coast of South America.

..... [1]

- (ii) Describe the movement of ocean currents in the Atlantic Ocean.

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..... [3]

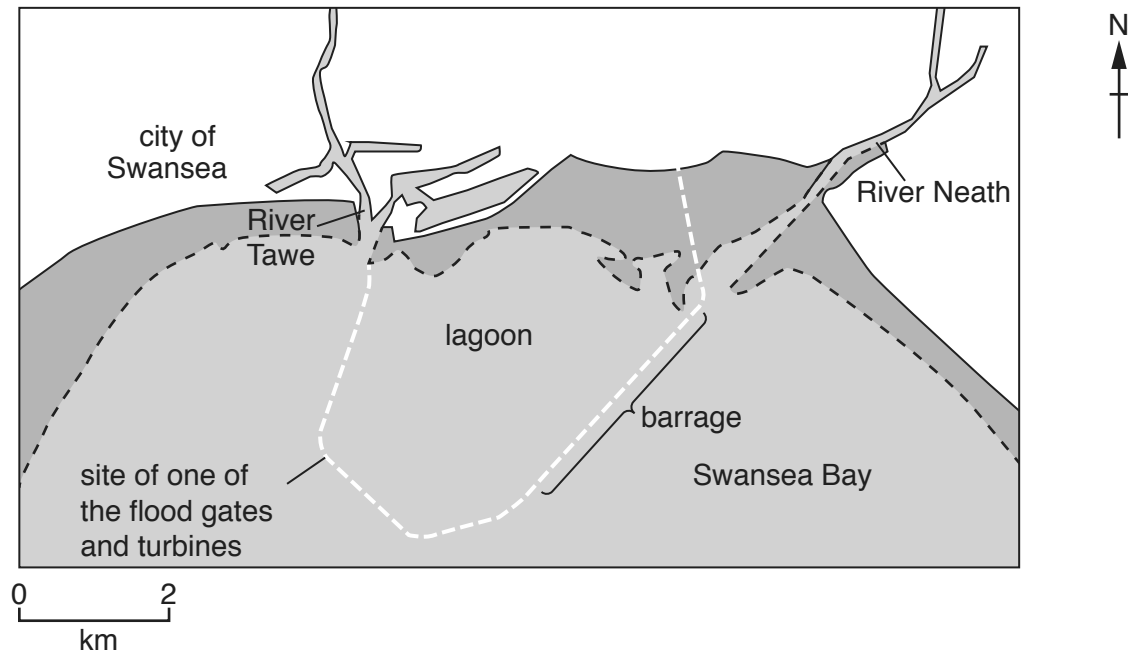
- (iii) Suggest how oil pollution from the Middle East can reach Antarctica.

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..... [2]

- (iv) Describe the effect of the Benguela current on the climate of the west coast of southern Africa.

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..... [2]

- (c) The information is about a proposed tidal barrage for generating electricity in Swansea Bay in the UK.

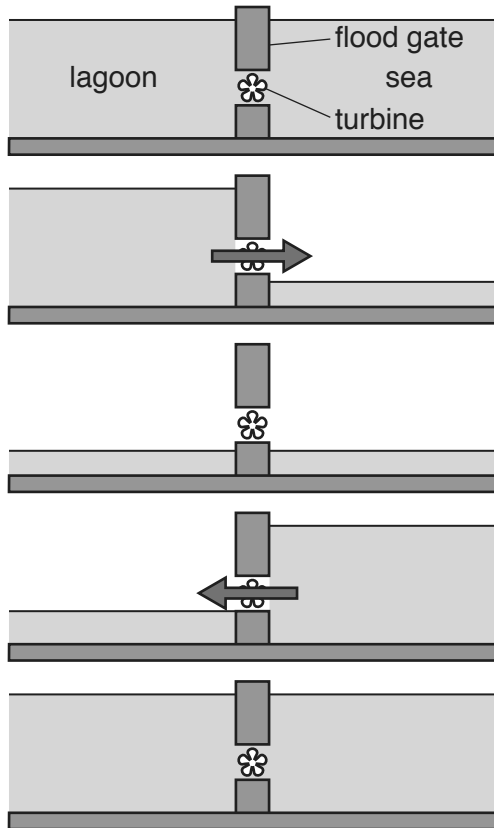


Swansea Bay has one of the world's largest tidal ranges, often reaching 10m. A 10km long barrage would be built 3km out to sea. The barrage would have 16 underwater turbines generating electricity on both the rising (incoming) and falling (outgoing) tide. Enough renewable power would be generated for 155 000 homes for 120 years.

An environmental impact assessment (EIA) is needed before the barrage can be built. The EIA will identify environmental impacts on coastal ecosystems and suggest ways to reduce them.

How it works

As the tide goes out, the flood gates in the barrage stay closed and the lagoon stays full. The flood gates are then opened to let the water out until water levels on each side of the barrage are the same. When the tide comes in the process is reversed.



starting point; high tide; lagoon full



tide goes out forming
a head of water in the lagoon



low tide; no head of water in the lagoon



tide comes in forming
a head of water outside the lagoon



return to starting point

(i) State the tidal range in Swansea Bay.

..... m [1]

(ii) Briefly describe how electricity will be generated at the barrage.

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..... [3]

- (iii) Explain why the barrage is a renewable method of generating electricity.

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..... [2]

- (iv) State how many homes will be provided with electricity by the barrage.

..... [1]

- (v) Suggest why it is important that the rivers Tawe and Neath still flow into the sea rather than into the lagoon.

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..... [3]

- (vi) Suggest why people are concerned about the environmental impact of the barrage.

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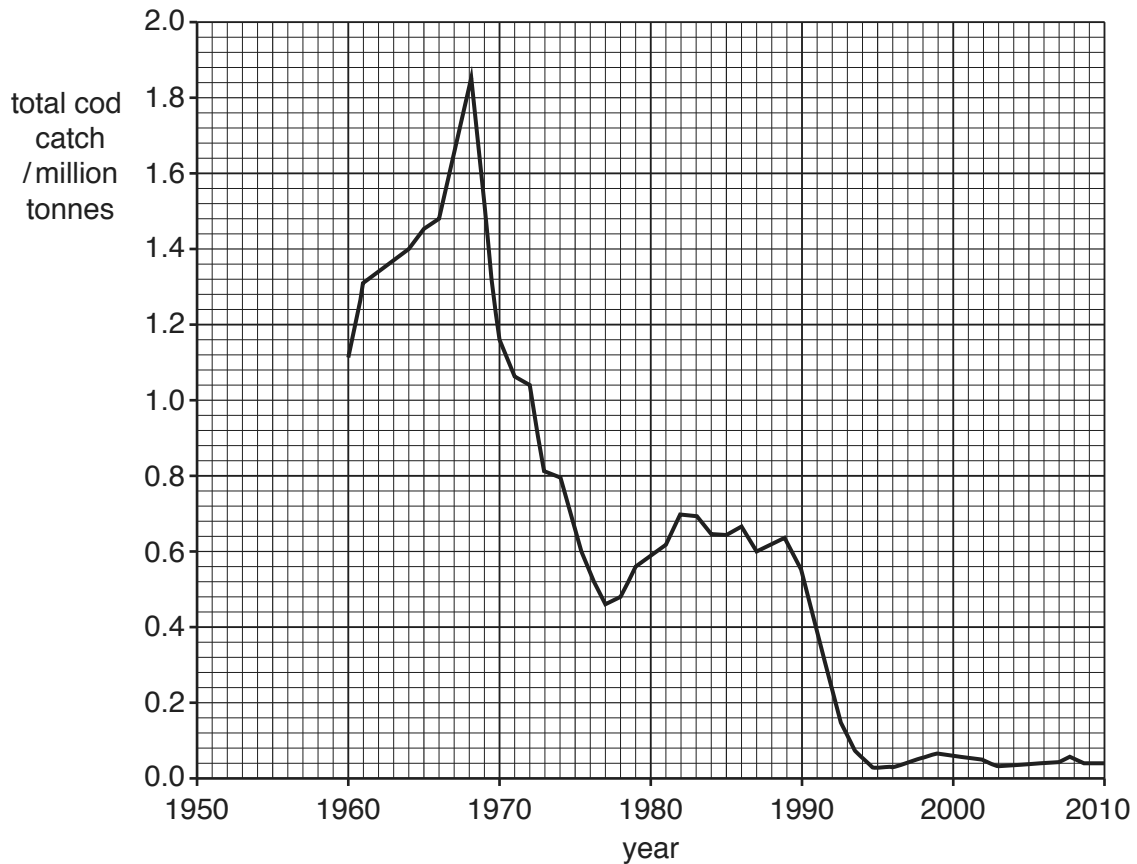
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- (d) The graph shows the cod catch in the Northwest Atlantic Ocean from 1950 to 2010. Cod are a species of fish.



- (i) Complete the graph using the data in the table.

year	total cod catch /million tonnes
1950	0.6
1955	0.9

[1]

- (ii) Describe the change in the cod catch in the Northwest Atlantic Ocean from 1960 to 1995.

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..... [3]

- (iii) Suggest **one** reason why the cod catch has remained low since 1995.

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..... [1]

- (iv) Cod are part of the Atlantic Ocean ecosystem.

Explain the meaning of the term *ecosystem*.

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..... [2]

- (e) 'Plastics cause greater damage to marine environments than raw sewage and heavy metals.'

How far do you agree with this statement? Give reasons for your answer.

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..... [6]

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